

Problem

A circuit has a transfer function

$$H(s) = \frac{3(s+4)}{(s+1)(s+2)^2}$$

Find the impulse response.

Step-by-step solution

Step 1 of 3

Consider the transfer function of a certain circuit is,

$$H(s) = \frac{3(s+4)}{(s+1)(s+2)^2}$$

Decompose $H(s)$ into simple terms using the residue method of partial fraction expansion.

$$\begin{aligned} H(s) &= \frac{3(s+4)}{(s+1)(s+2)^2} \\ &= \frac{A}{(s+1)} + \frac{B}{(s+2)^2} + \frac{C}{s+2} \end{aligned}$$

Here,

A , B , and C are constants.

Step 2 of 3

Calculate the constant value A using residue method.

$$\begin{aligned}
 A &= (s+1)H(s)\Big|_{s=-1} \\
 &= (s+1)\left(\frac{3(s+4)}{(s+1)(s+2)^2}\right)\Big|_{s=-1} \\
 &= \left(\frac{3(-1+4)}{(-1+2)^2}\right) \\
 &= 9
 \end{aligned}$$

Calculate the constant value B using residue method.

$$\begin{aligned}
 B &= (s+2)^2 H(s)\Big|_{s=-2} \\
 &= (s+2)^2 \left(\frac{3(s+4)}{(s+1)(s+2)^2}\right)\Big|_{s=-2} \\
 &= \left(\frac{3(-2+4)}{-2+1}\right)\Big|_{s=-2} \\
 &= -6
 \end{aligned}$$

Calculate the constant value C using residue method.

$$\begin{aligned}
 C &= \frac{d}{ds}(s+2)^2 H(s)\Big|_{s=-2} \\
 &= \frac{d}{ds}\left[(s+2)^2 \left(\frac{3(s+4)}{(s+1)(s+2)^2}\right)\right]\Big|_{s=-2} \\
 &= 3 \frac{d}{ds}\left(\frac{s+4}{s+1}\right)\Big|_{s=-2} \\
 &= 3 \frac{(s+1)(1) - (s+4)(1)}{(s+1)^2}\Big|_{s=-2} \\
 &= (3)\left(\frac{-1-2}{1}\right) \\
 &= -9
 \end{aligned}$$

Substitute obtained A , B , and C values in equation of $H(s)$.

$$H(s) = \frac{9}{(s+1)} + \frac{-6}{(s+2)^2} + \frac{-9}{s+2}$$

Definition of impulse response:

It is the output response of a circuit when the input is an impulse function $\delta(t)$. The Laplace transform of an impulse function $\delta(t)$ is 1 ($\mathcal{L}(\delta(t))=1$). So, the impulse response is same as the inverse transform of the transfer function, $H(s)$.

Calculation of impulse response, which is inverse transform of the transfer function, $H(s)$.

$$\begin{aligned}
 h(t) &= \mathcal{L}^{-1}\{H(s)\} \\
 &= \mathcal{L}^{-1}\left\{\frac{9}{(s+1)} + \frac{-6}{(s+2)^2} + \frac{-9}{s+2}\right\} \\
 &= \mathcal{L}^{-1}\left\{\frac{9}{(s+1)}\right\} + \mathcal{L}^{-1}\left\{\frac{-6}{(s+2)^2}\right\} + \mathcal{L}^{-1}\left\{\frac{-9}{s+2}\right\} \quad \left(\because e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}\right. \\
 &\quad \left.te^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{(s+a)^2}\right) \\
 &= 9e^{-t}u(t) - 6te^{-2t}u(t) - 9e^{-2t}u(t) \\
 &= [9e^{-t} - 6te^{-2t} - 9e^{-2t}]u(t)
 \end{aligned}$$

Thus, the impulse response of the circuit is $\boxed{[9e^{-t} - 6te^{-2t} - 9e^{-2t}]u(t)}$.